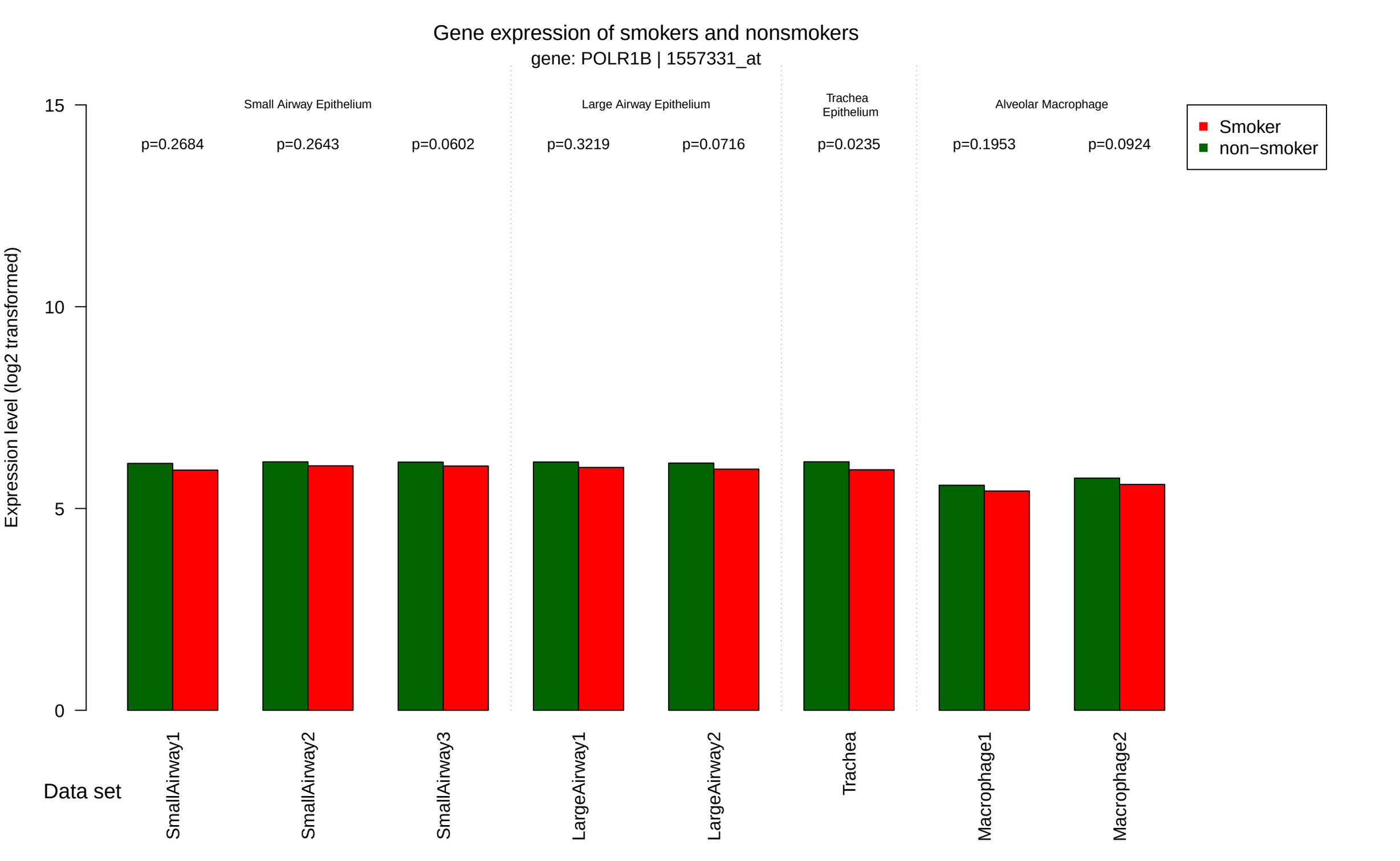
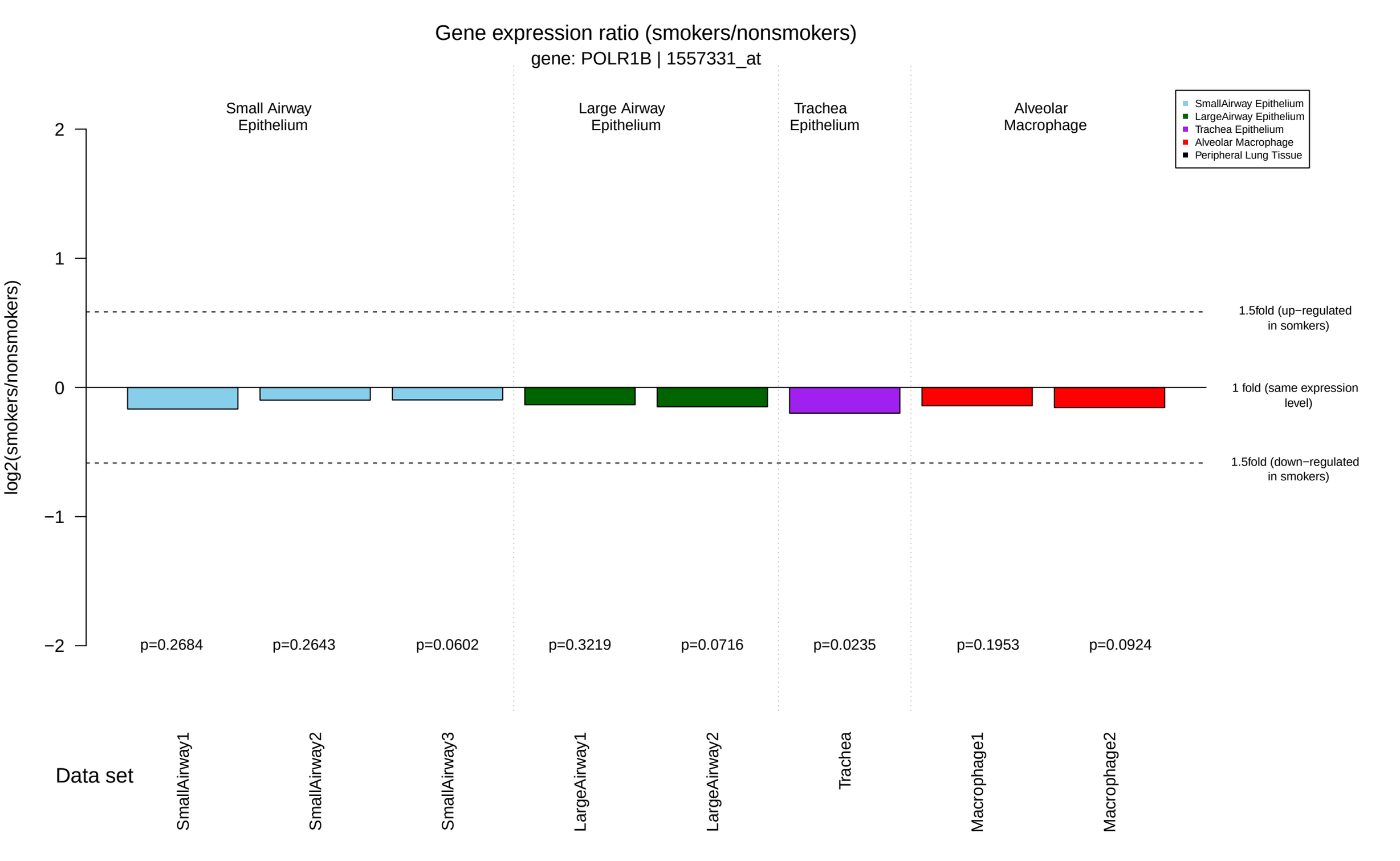
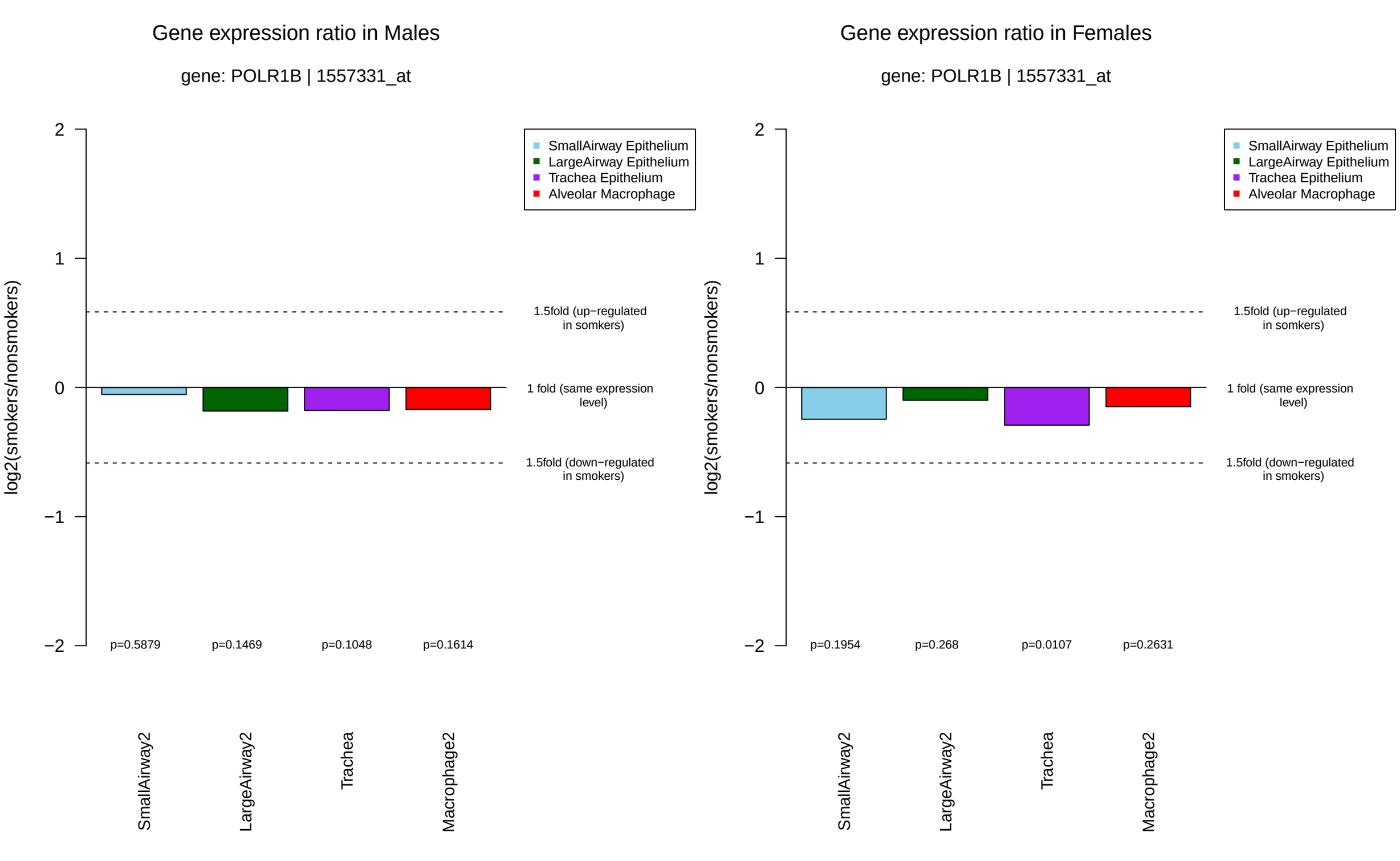
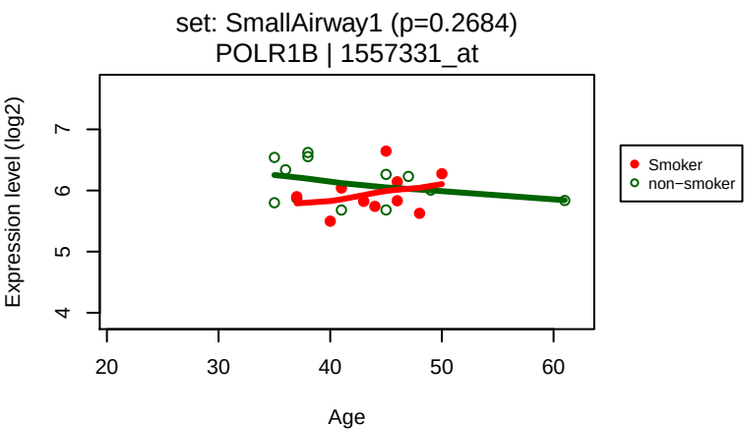
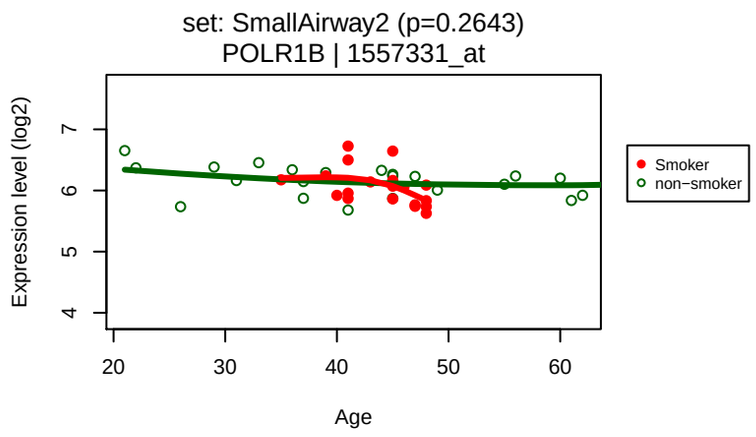
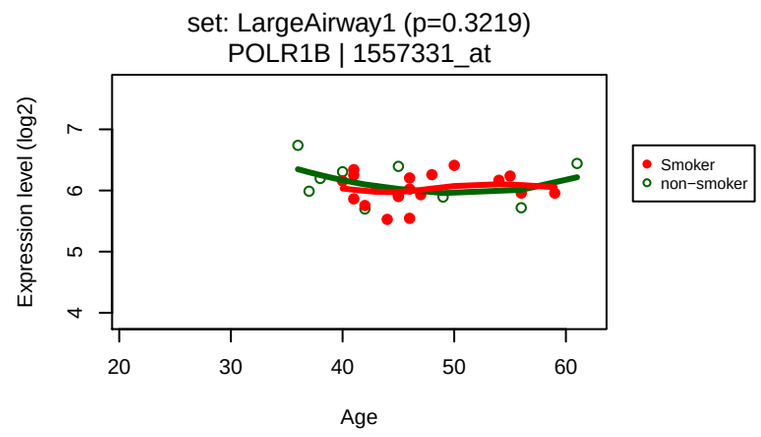

Gene expression of smokers and nonsmokers
gene: POLR1B | 1557331_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2684
p=0.2643
p=0.0602
p=0.3219
p=0.0716
p=0.0235
p=0.1953
p=0.0924
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

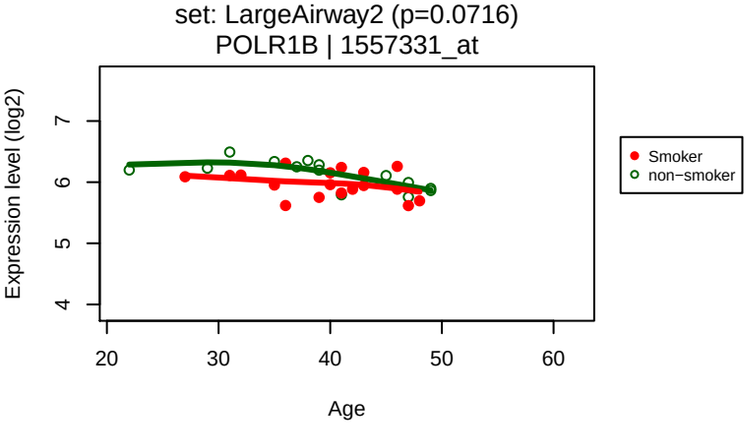

Gene expression ratio (smokers/nonsmokers)
gene: POLR1B | 1557331_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2684
p=0.2643
p=0.0602
p=0.3219
p=0.0716
p=0.0235
p=0.1953
p=0.0924
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

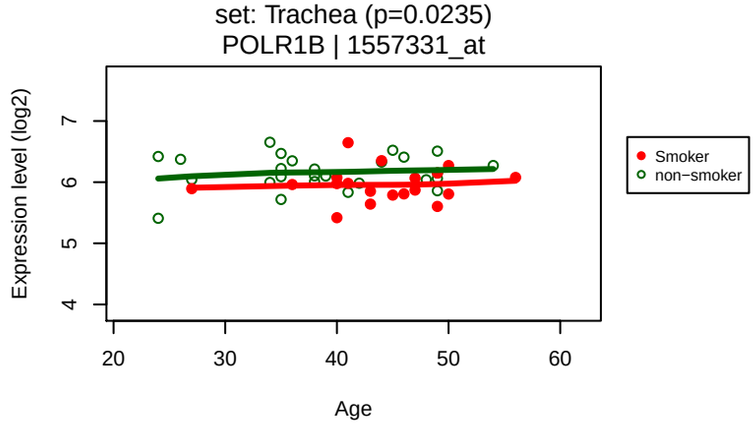

Gene expression ratio in Males
gene: POLR1B | 1557331_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5879
p=0.1469
p=0.1048
p=0.1614
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: POLR1B | 1557331_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1954
p=0.268
p=0.0107
p=0.2631
SmallAirway2
LargeAirway2
Trachea
Macrophage2

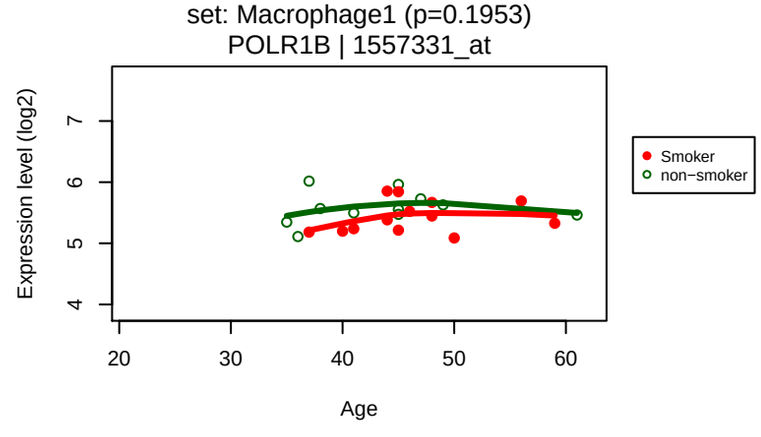

set: SmallAirway1 (p=0.2684)
POLR1B | 1557331_at
Expression level (log2)
Age
Smoker
non−smoker

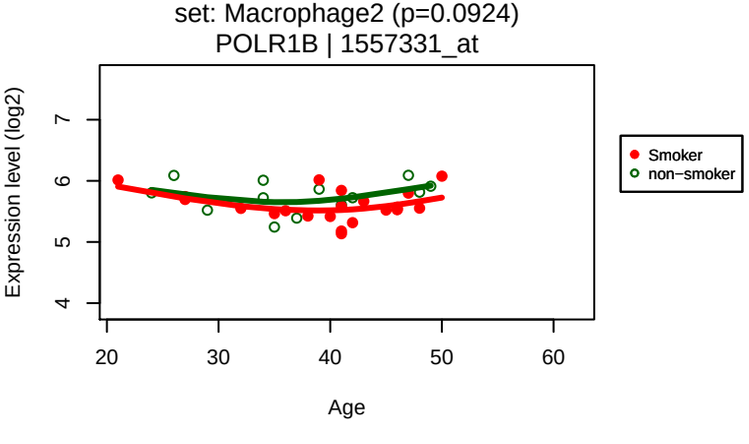

set: SmallAirway2 (p=0.2643)
POLR1B | 1557331_at
Expression level (log2)
Age
Smoker
non−smoker

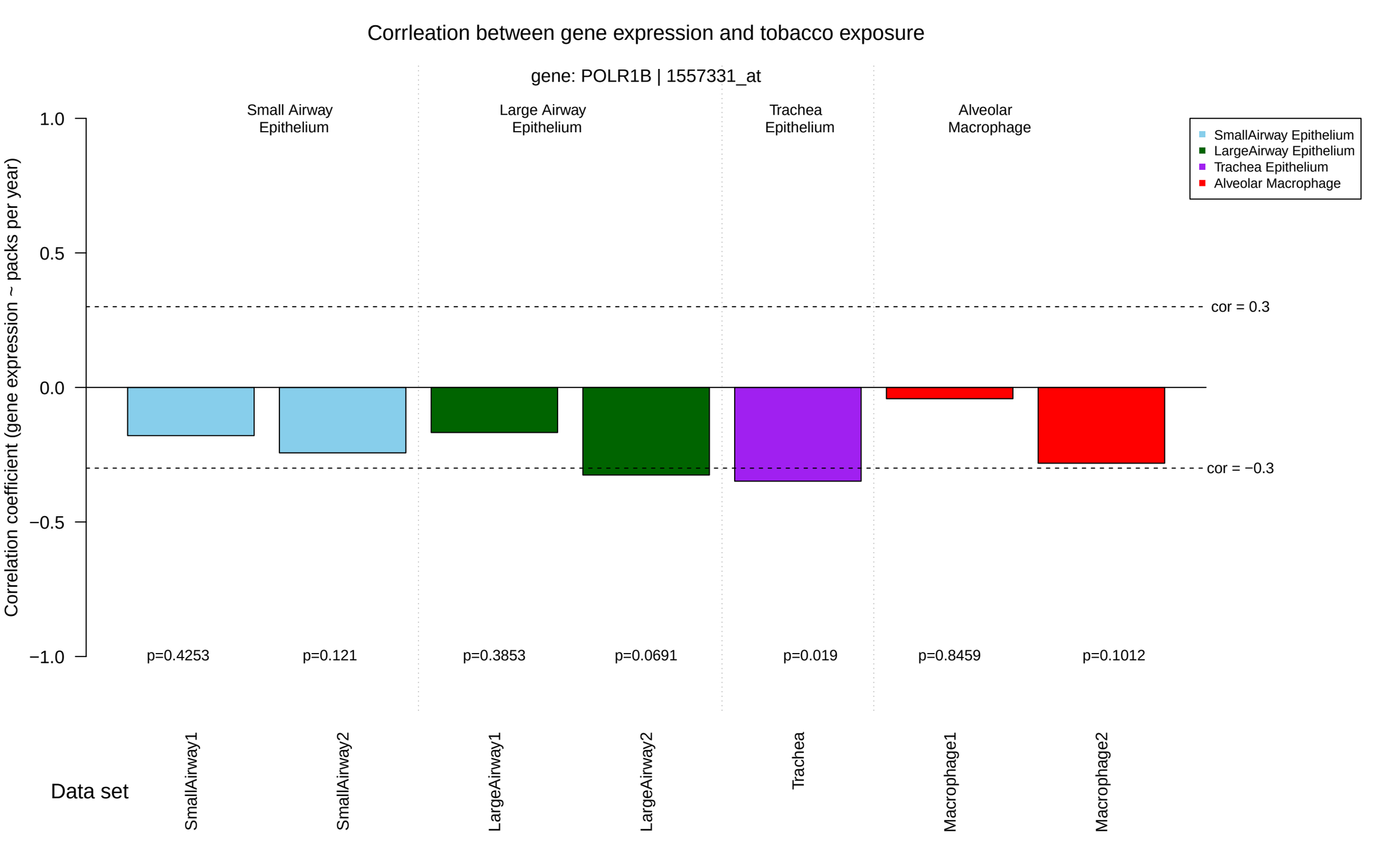

set: LargeAirway1 (p=0.3219)
POLR1B | 1557331_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0716)
POLR1B | 1557331_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0235)
POLR1B | 1557331_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1953)
POLR1B | 1557331_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0924)
POLR1B | 1557331_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: POLR1B | 1557331_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4253
p=0.121
p=0.3853
p=0.0691
p=0.019
p=0.8459
p=0.1012
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2